

Charles T. Marx

CURRICULUM VITAE - JULY 2019

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<https://charliemarx.github.io/> 🌐

EDUCATION

Haverford College

2016 – PRESENT

B.S. Computer Science Candidate

B.S. Mathematics Candidate

Minor in Statistics Candidate

GPA: 3.94/4.0

AWARDS & HONORS

Beckman Scholars Award

2018

National undergraduate scholarship for excellence in research.

Brian Kovaric Award Finalist

2017

Faculty-nominated scholarship for one outstanding first-year student.

Farmers Insurance Standout Student of the Year Award

2016

Regional award for academic excellence and community engagement awarded to one student each year.

ACADEMIC POSITIONS

Harvard University

2019 – PRESENT

Research Assistant, Department of Computer Science

Developed methods for finding equivalent models for recidivism prediction and theoretical work to develop their implications.

Advised by: Burk Ustun and Flavio Calmon

Haverford College

2017 – 2019

Research Assistant, Department of Computer Science

Machine learning research spanning adversarial learning paradigms, disentangled representations, and fairness and interpretability of black box models.

Advised by: Sorelle Friedler

Haverford College

2017 – 2018

Research Assistant, Department of Chemistry

Data mining and machine learning in environmental chemistry. Published work using machine learning to assist in rapid identification of oil in the environment.

Advised by: Helen White

Prince William Sound Science Center

2018

Summer Research Intern, Herring Research and Monitoring Project

Machine learning analysis of oceanographic datasets such as aerial herring surveys and sonar data.

Advised by: Scott Pegau

Haverford College

2017

Research Assistant, Digital Scholarship

Used recurrent neural networks for handwriting recognition in a dataset of 100,000 handwritten Spanish texts.

Advised by: Andrew Janco

PUBLICATIONS

[Disentangling Influence: Using Disentangled Representations to Audit Model Predictions](#)

Charles T. Marx, Richard Lanas Phillips, Sorelle A. Friedler, Carlos Scheidegger, Suresh Venkatasubramanian
Conference on Neural Information Processing Systems (NeurIPS), 2019 – *In review*

[Examining Inputs of Biogenic and Oil-Derived Hydrocarbons in Surface Waters Following the Deepwater Horizon Oil Spill](#)

Helen K. White, **Charles T. Marx**, David L. Valentine, Charles Sharpless, Christoph Aeppli, Kelsey M. Gosselin, Veronika Kivenson, Rachel M. Liu, Robert K. Nelson, Sean P. Sylva, Christopher M. Reddy
ACS Earth and Space Chemistry, 2019 – *Accepted*

[Rapid Identification of Deepwater Horizon Oil Residues Using X-Ray Fluorescence](#)

Anna P. M. Michel, Alexandra E. Morrison, **Charles T. Marx**, Helen K. White
Environmental Science & Technology Letters, 2018 – *Accepted*

ORAL PRESENTATIONS

Simons Institute for the Theory of Computing

2019

Berkeley, CA

Disentangling Influence (45 minute talk)

Fairness, Accountability and Transparency in Machine Learning

2018

New York, NY

Auditing Black Box Models (60 minute tutorial)

Gulf of Mexico Oil Spill & Ecosystem Science Conference

2018

New Orleans, LA

Examining Inputs of Biogenic and Oil-Derived Hydrocarbons (15 minute talk)

POSTER PRESENTATIONS

Beckman Regional Symposium

2018

New York, NY

Deep Learning Methods for Mitigating Social Discrimination

KINSC Undergraduate Research Symposium

2017

Haverford, PA

Examining Biogenic and Oil-Derived Hydrocarbons in Surface Waters After the Deepwater Horizon Oil Spill

TRAVEL GRANTS

Summer Cluster on Fairness: Simons Institute for the Theory of Computing
Berkeley, CA (2019)

Fairness, Accountability and Transparency in Machine Learning (FAT*)
New York, NY – Atlanta, GA (2018, 2019)

International Conference on Knowledge Discovery & Data Mining (KDD)
London, UK (2018)

Prince William Sound Auklet Research Charter
Valdez, AK (2018)

Gulf of Mexico Oil Spill & Ecosystem Science Conference
New Orleans, LA (2017)

Functional Ecological Genomics Workshop
Lake Ariel, PA (2017)

ACADEMIC SERVICE

Computer Science Faculty Search Committee 2018 – 2019
Member of search committee tasked with hiring a tenure-track faculty member for the computer science department.

Perovskite Solar Cell Chemistry Group 2017
An extracurricular chemistry lab to design perovskite solar cells for the introductory lab curriculum.

FIELD EXPERIENCE

Auklet Research Charter 2018
Prince William Sound, AK
Collected oil residues from island shorelines impacted by the 1989 Exxon Valdez oil spill and the 1964 Alaska Earthquake while at sea for one week.

Gulf of Mexico Oil Survey 2017
Pensacola, FL to Grand Isle, LA
Surveyed and collected oil samples from Gulf of Mexico beaches impacted by the Deep-water Horizon oil spill while traveling along the coast for one week.

EXTRACURRICULAR ACTIVITIES

Haverford College Men's Club Soccer 2018 – 2019
Captain.

Haverford College Outdoors Club Leader 2017 – 2018
Designed and led week-long backpacking trips for students at Haverford College.

Furniture Forward 2015 – 2017
Leader of organization to redistribute furniture to families transitioning out of homelessness.